bom.xlsx — sheet BOM

No	Component	Layer	Mass [g/cell]	Mass [kg/kWh]	Notes / source
1	LNMO spinel active (positive)	Positive electrode	17.3088	0.8198	96 wt% of positive layer (18.0300 g); literature-default electrode formulation, not simulated
2	Graphite active (negative)	Negative electrode	13.2161	0.626	95 wt% of negative layer (13.9117 g); literature-default electrode formulation, not simulated
3	Carbon black conductive additive	Both electrodes	0.778	0.0368	2 wt% positive + 3 wt% negative (literature default)
4	PVDF binder	Positive electrode	0.3606	0.0171	2 wt% of positive layer (literature default)
5	SBR/CMC binder	Negative electrode	0.2782	0.0132	2 wt% of negative layer (literature default)
6	Separator (polyolefin, 12 um)	Separator	0.2593	0.0123	solid fraction 0.53, density 397 kg/m3 [Chen2020.py:308,306]
7	Aluminium foil (16 um)	Positive current collector	4.4366	0.2101	density 2700 kg/m3 [Chen2020.py:261]
8	Copper foil (12 um)	Negative current collector	11.0423	0.523	density 8960 kg/m3 [Chen2020.py:260]
9	Electrolyte (2.0 M EC-lean, LiPF6)	Pore volume fill	6.5305	0.3093	pore volume 5.4421 mL x 1.2 g/mL default density; EXCLUDED from contract mass (electrolyte_included =False)
	TOTAL (dry, contract basis)		47.68	2.2584	sum of rows 1-8; equals r10_V27_combo_energy.json mass_kg (47.6800 g)
	TOTAL (wet, incl. electrolyte)		54.2105	2.5677	dry + row 9; electrolyte density 1.2 g/mL literature default
	Energy basis		21.11246886967024		r10_V27_combo_energy.json energy_wh [Wh]; kg/kWh = g/(Wh/1000)